MIL-C-41820D

1 March 1976
SUPERSEDING
MIL-C-41820C
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MILITARY SPECIFICATION

CLOTH, GABARDINE, POLYESTER AND RAYON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers polyester and rayon spun yarn gabardine cloth (see 6.1).

1.2 Classification. The cloth shall be of the following types and classes as specified (see 6.2):

Type I	-	6.0 ounce
Type II	-	6.4 ounce
Type III	-	8.0 ounce
Class 1	-	Bleached White
Class 2	-	Dyed

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

V-T-285	-	Thread, Polyester
PPP-P-1133	-	Packaging and Packing of Synthetic Fiber Fabrics

STANDARDS

FEDERAL

FED-STD-4	-	Glossary of Fabric Imperfections
FED-STD-191	-	Textile Test Methods

FSC 8305

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MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies may be obtained without charge from the Federal Trade Commission, Washington, DC 20580.)

3. REQUIREMENTS

3.1 Standard sample. When a standard sample is available (see 6.3), the cloth shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced.

3.2 Material.

3.2.1 Fiber blend.

3.2.1.1 Polyester fiber. The fiber shall be polyethylene glycol terephthalate. The use of optically brightened fiber for type I and type III, class 1 is permitted. Unless otherwise specified, the use of optically brightened fiber for type II and type III, class 2 is prohibited.

3.2.1.2 Rayon fiber. The fiber shall be rayon staple. The use of rayon waste of any kind is prohibited.

* 3.2.2 Yarn. The warp and filling yarn shall be a blend of polyester staple fiber and rayon staple fiber of such proportions as to analyze not less than 65 percent nor more than 75 percent polyester fiber when tested as specified in 4.4. The yarn for all types shall be spun and twisted into 2 ply yarn.

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3.3 Color.

3.3.1 Class 1, bleached. The color of the finished type I and type III, class 1, cloth shall be a bleached white conforming to the standard sample when tested as specified. All cloth within any one contract shall be processed for whiteness in the same manner. The color shall be obtained with a chemical bleach, preferably with hydrogen peroxide or peroxygen chemicals. The chemical bleach may be supplemented with a blue-violet fluorescing optical brightener.

* 3.3.2 Class 2, dyed. The color of the cloth shall be as specified and shall match the standard sample (see 6.2 and 6.3).

* 3.3.2.1 Labile sulfur. The use of dyes and compounds containing elementary sulfur capable of oxidation to sulfuric acid is prohibited. The dyestuff shall be chosen and applied so that the dyed and finished cloth shall contain no more labile sulfur than shown by the standard sample when tested as specified in 4.4. When a standard sample is not available the dyed and finished cloth shall show not more than a slight trace of labile sulfur as defined in the test method when tested as specified in 4.4.

* 3.3.3 Matching. The color of class 2 cloth shall match the standard sample under artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the standard sample under incandescent lamp-light at 2800° Kelvin. The color of the class 1 cloth shall match the standard sample under 7500° Kelvin light and also under 7500° Kelvin light with ultra-violet component added.

3.3.4 Colorfastness. The dyed finished cloth shall show fastness to light, laundering, perspiration and crocking equal to or better than the standard sample when tested as specified in 4.4. When no standard sample is available, or when the standard sample is not referenced for colorfastness, the dyed cloth shall show "good" fastness to light, laundering and perspiration, and no more crocking than Munsell value 8.5. Type I cloth and type III, class 1 cloth shall show no more discoloration in light than that of the standard sample when tested as specified in 4.4.

3.4 Physical requirements. The finished cloth shall conform to the requirements shown in table I when tested as specified in 4.4.

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TABLE I. Physical requirements

	Weight, ounces per square yard (minimum)	Yarns per inch (minimum)		Breaking strength, pounds (minimum)		Weave
		Warp	Filling	Warp	Filling	
Type I	6.0	110	52	210	90	2 up-1 down right twill
Type II	6.4	110	62	210	105	2 up-1 down right twill
Type III	8.0	92	42	280	120	2 up-1 down right twill

3.5 Width. The minimum width (inclusive of the selvages) of the finished cloth shall be as specified (see 6.2).

3.6 Nonfibrous materials.

3.6.1 Type I and type III, class 1. The starch and protein content including chloroform-soluble and water-soluble material after bleaching shall not exceed 2.0 percent when tested as specified in 4.4.

3.6.2 Type II and type III, class 2 The starch and protein content, including chloroform-soluble and water-soluble material after dyeing and prior to the resin finish, shall not exceed 2.0 percent when tested as specified in 4.4. The resin finish applied to the dyed cloth shall not exceed 12.0 percent.

3.7 Finish.

3.7.1 Type I and type III, class 1. The type I and type III class 1 cloth shall be scoured, heat set and bleached to provide a finish equal in character to that of the standard sample and meet the requirements specified herein. A resin finish may be used to control shrinkage. Resins other than those approved for use are prohibited (see 6.7).

3.7.2 Type II and type III, class 2. The type II and type III class 2 cloth shall be scoured, heat set, dyed and resin treated, and washed to provide a finish equal in character to that of the standard sample (see 6.7). The resin treated cloth shall be processed at such a time and temperature cycle that will insure adequate curing of the resin, and shall be washed thoroughly to remove all traces of uncured resin.

3.8 pH. The pH value of the water extract of the finished cloth shall be not less than 5.0 nor more than 8.5 when tested as specified in 4.4.

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3.9 Shrinkage.

3.9.1 Type I and type III. The cloth shall be preshrunk so that it shall not shrink more than 3.5 percent in the warp and 2.0 percent in the filling when tested as specified in 4.4. The preshrinking process shall not be indentified by name or trademark either on the cloth or on the ticket or package.

3.9.2 Type II. The cloth shall be preshrunk so that the cloth shall not shrink more than 2.5 percent in the warp and 1.5 percent in the filling when tested as specified in 4.4. The preshrinking process shall not be identified by name or trademark either on the cloth or on the ticket or package.

3.10 Seam efficiency. The cloth shall have a seam efficiency of not less than 85 percent when tested as specified in 4.4.

* 3.11 Length and put-up. Unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths of not less than 40 yards. Each length shall be put-up on rolls as specified in PPP-P-1133.

3.12 Fiber identification. Each roll shall be labeled, ticketed or for fiber content in accordance with the Textile Fiber Products Identification Act.

3.13 Workmanship. The finished cloth shall conform to the quality established by this specification. The demerit points per 100 square yards when calculated as specified in section 4 shall not exceed the established maximum point value.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

* 4.2 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

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4.2.2 Examination of the end item. Defects found during examination shall be classified in accordance with 4.2.2.1 thru 4.2.2.4.

* 4.2.2.1 Yard-by-yard examination. Each roll in the sample shall be examined on the face side only. When the total yardage in the roll does not exceed 100 yards, the entire yardage in the roll shall be examined. When the total yardage in the roll exceeds 100 yards, only 100 yards shall be examined. All defects as defined in Section III of FED-STD-4 which are clearly noticeable at normal inspection distance (3 feet) shall be scored and assigned demerit points as listed in 4.2.2.1.1 except that only those slubs and knots which exceed the limits shown on Sears Fabric Defect Scales (see 6.6) D for slubs and C for knots, shall be scored. No linear yard or (increment of one yard on the measuring device of the inspection machine) from any one roll within the sample, shall be penalized more than 4 points. The sample unit for this examination shall be one linear yard. The sample size shall be 20 rolls selected from 20 containers. The lot shall be unacceptable if the points per 100 square yards exceed the following:

35.0 points for dyed cloth
30.0 points for white cloth

The lot shall be unacceptable if two or more individual rolls exceed the following:

53.0 points for dyed cloth
45.0 points for white cloth

If one roll exceeds the point level, a second sample of 20 rolls shall be selected for individual roll quality examination. The lot shall be unacceptable if one or more rolls in the second sample exceeds the point level. Point computation for the lot quality and individual roll quality shall be as follows:

$$\frac{\text{Total points scored in sample} \times 3600}{\text{Contracted width of cloth (inches)} \times \text{Total yards inspected}} = \text{Points per 100 square yards}$$

4.2.2.1.1 Demerit points. Demerit points shall be assigned as follows:

For defects 3 inches or less in any dimension.	- one point
For defects exceeding 3 inches, but not exceeding 6 inches in any dimension.	- two points
For defects exceeding 6 inches, but not exceeding 9 inches in any dimension.	- three points
For defects exceeding 9 inches in any dimension.	- four points

NOTE: The following defects when present shall be scored four points for each yard in which they occur:

Width of cloth less than specified.
Poor dye penetration, streaky or cloudy.
Excessive neppiness.

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4.2.2.2 Examination for length.

4.2.2.2.1 Individual rolls. During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or more than two yards below the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective in respect to length.

4.2.2.2.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of lengths marked on the ticket.

* 4.2.2.3 Examination for shade and finish. Each roll in the sample shall be examined visually for shade match (see 3.3.3) and finish (see 3.7). A roll shall be unacceptable if it fails to meet the requirements for shade match and finish. The sample unit shall be a 4-inch by 20-inch swatch of the cloth. A sample unit shall be drawn from each roll in the lot.

4.2.2.4 Examination for identification of preshrinkage process and compliance with Textile Fiber Products Identification Act. During the yard-by-yard examination each roll in the sample shall be examined for these defects. The lot shall be unacceptable if two or more rolls in the sample contain identification of the preshrinkage process by name or trademark on the cloth or ticket, or are not labelled or ticketed in accordance with the Textile Fiber Products Identification Act.

4.3 Examination of preparation for delivery requirements. An examination shall be made in accordance with the provisions of PPP-P-1133 to determine that packaging, packing and marking comply with the section 5 requirements.

4.4 Testing of the end item. The methods of testing specified in FED-STD-191, wherever applicable and as listed in table II, shall be followed. The physical and chemical values specified in section 3 apply to the results of the determinations made on a sample unit for test purposes as specified in the applicable test method. All test reports shall contain the individual values utilized in expressing the final result. The sample unit shall be 4-1/2 continuous yards full width of the finished cloth, and 1/4 yard full width of cloth prior to resin treatment. The sample size (number of sample units) shall be as shown in the table below. The lot size shall be expressed in units of 1 yard. The lot shall be unacceptable if one or more sample units fail to meet any test requirement specified.

Lot size (yards)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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TABLE II. Test methods

Characteristic	Requirement paragraph	Test method
Polyester fiber identification	3.2.1.1	1/
Rayon fiber identification	3.2.1.2	1/
Polyester fiber content:		
Warp	3.2.2	4.4.1 1/
Filling	3.2.2	4.4.1 1/
Yarn ply	3.2.2	1/
Bleaching agents/brighteners	3.3.1	1/
Labile sulfur (prior to treatment)	3.3.2.1	2020
Colorfastness to:		
Light	3.3.4	5660
Laundering	3.3.4	5614
Perspiration	3.3.4	5680
Crocking	3.3.4	5651
Discoloration (type I and type III, class 1)	3.3.4	2/
Weight oz./sq. yd.	3.4	5041
Yarns per inch		
Warp	3.4	5050
Filling	3.4	5050
Breaking strength		
Warp	3.4	5100
Filling	3.4	5100
Weave	3.4	Visual 3/
Nonfibrous material content		
Before finishing	3.6.1 and 3.6.2	2611
After finishing (type II and type III, class 2)	3.6.2	1/

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TABLE II. Test methods (cont'd)

Characteristic	Requirement paragraph	Test method
Finish (types I, II and III)		
Scoured	3.7.1 and 3.7.2	1/
Heat set	3.7.1 and 3.7.2	1/
Resin treated (type II and type III, class 2)	3.7.2	1/
Resin identification	3.7.2	1/
pH	3.8	2811
Shrinkage in laundering (type I and III)	3.9.1	5552
Shrinkage in laundering (wool formula with tumble drying) (type II)	3.9.2	5556
Seam efficiency	3.10	5110 4/

1/ Unless otherwise specified a certificate of compliance shall be submitted and will be acceptable for the stated requirements.

2/ For type I and type III, class 1 cloth, Method 5660 shall be used except that the exposure time shall be 20 standard fading hours and the exposed samples shall be allowed to lie in the dark at room temperature for a minimum of 48 hours before comparison. Any specimen showing discoloration less than or equal to that shown by the standard sample shall be considered satisfactory. Any specimen showing discoloration greater than that of the standard sample shall be considered unsatisfactory. One determination shall be made for each sample unit and the result reported as "pass" or "fail".

3/ One determination per sample unit and results reported as "pass" or "fail".

4/ The needle size shall be .040 inch ± .001 inch. The needle thread shall be size B polyester yarn and the looper thread shall be size A polyester yarn in accordance with type I of V-T-285.

* 4.4.1 Polyester fiber content.

4.4.1.1 Test specimen. The specimen shall be approximately 2 grams made up of yarns taken from the cloth sample and cut into approximately 1/4 inch lengths.

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4.4.1.2 Apparatus, reagents, method cited.

Apparatus as described in Method 2611.
Sintered glass filter crucible of coarse porosity.
Hydrometer.
Sulfuric acid, 75 percent (density 1.67) prepared by pouring first 350 cubic centimeters (cc.) of water in a round-bottom flask, then adding 680 cc. of concentrated sulfuric acid slowly, with cooling under the tap. The density of the solution is adjusted to 1.67 using a hydrometer sulfuric acid, dilute (100 cc. of concentrated sulfuric acid added to 1900 cc. of water).

4.4.1.3 Procedure. The nonfibrous material shall be removed as specified in Method 2611. The desized specimen shall be dried to a constant weight, cooled in a desiccator and weighed. The speciman shall then be immersed in 200 cc. of 75 percent solution of sulfuric acid for 1 hour with intermittent stirring. The residual fibers shall be transferred to the glass crucible and the excess acid solution removed by suction. The residual fibers shall be washed on the fiber filter successively with dilute sulfuric acid, water, dilute ammonia solution (soak for ten minutes) and again with water (soak for ten minutes). After each separate washing, the crucible is drained by suction. The residue shall be dried to a constant weight in the air oven at 105° to 110°C. This is the "weight of the dry polyester fiber' in the specimen.

4.4.1.4 Result. The percentage of polyester fiber shall be based on the following equation:

$$\text{Polyester fiber, percent} = \frac{\text{Weight of dry polyester fiber}}{\text{Weight of dry desized specimen}} \times 100$$

100 - percent polyester = percent rayon

Two specimens shall be tested from each sample unit and the average percent polyester fiber content and the average percent rayon of the two specimens shall be reported to the nearest 1.0 percent.

5. PACKAGING

* 5.1 Put-up and packaging. Put-up and packaging shall be level A or C as specified (see 6.2).

5.1.1 Levels A and C. The cloth, put-up as specified, shall be packaged in accordance with the applicable requirements of PPP-P-1133.

5.2 Packing. Packing shall be Level A, B, or C as specified (see 6.2 and 6.2.1).

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5.2.1 Levels A, B and C. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1133.

5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with PPP-P-1133.

6. NOTES

6.1 Intended use. The cloth is intended for use in the manufacture of clothing items for personnel of the Department of Defense. Specific uses are:

(a) Type I - Women's Summer White Uniforms
- Uniform, Woman's Dress; White (Marine Corps Officers)
- Jacket, Man's; Mess; White (Marine Corps Officers)
- Coat, Man's White; Dress (Marine Corps Officers)
- Jacket, Woman's; Mess Dress (Marine Corps Officers)
- Trousers, Men's; Dress (White) (Marine Corps Officers)

(b) Type II - Cap, Field, Hot Weather

(c) Type III - Coveralls, Mechanics (basic cloth)
- Uniform; Woman's; Dress; White (Marine Corps Officers)
- Coat, Man's: White; Dress (Marine Corps Officers)
- Jacket, Man's: Mess; White (Marine Corps Officers)
- Trousers, Men's: Dress; White (Marine Corps Officers)

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type and class required (see 1.2).
(c) Color of cloth required (see 3.3).
(d) Width required (see 3.5).
(e) Length of roll required if other than specified (see 3.11).
(f) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2.1 In the preparation of contracts or orders, it must be noted that various options, choices and alternatives, as indicated in PPP-P-1133, may be exercised in the preparation for delivery of this cloth.

6.3 Standard sample. For access to standard sample, address the procuring office issuing the invitation for bids.

6.4 A suggested but not mandatory dye stuff formulation for Olive Green 106 is as follows:

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Polyester component

Dispersed Blue 62
Dispersed Blue 61
Dispersed Yellow 54
Dispersed Orange 25
Shaded with dispersed Red 60

Rayon component.

Vat Green 8, No. 71050
Vat Black 25, No. 69525
Vat Green 3, No. 69500
Shaded with Vat Yellow 2, No. 67300

6.5 Sears Fabric Defect Scales are available from Sears, Roebuck and Company, Department 733Q, Sears Defect Replica Kit, Sears Tower, Chicago, IL 60684.

6.6 Types I, II and III. To meet shrinkage requirements (see 3.9), it may be necessary to stabilize the cloth with resin. Attention is brought to the finish requirements of 3.7.1 and nonfibrous material requirements of 3.6.1 when resins are used. Approval of the resins for utilization under this specification is the responsibility of the U.S. Army Natick Research and Development Command, Natick, MA 01760, and is based on extensive tests, including those for human toxicity which are not set forth in this specification. Because of the time necessary to conduct full evaluations, only those treatments approved and so listed in the invitation for bids or request for proposal shall be considered acceptable for the related procurement.

* 6.7 The standard samples for type I and type II cloths were woven using 40/2 for both the warp and filling yarn.

6.8 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Custodians:

Army - GL
Navy - SA
Air Force - 11

Review activities:

Army - MD
Navy - MC
Air Force - 45, 82
DSA - CT

Preparing activity:

Army - GL

Project No. 8305-0097

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DOCUMENT IDENTIFIER AND TITLE
- MIL-C-41820D - CLOTH, GABARDINE, POLYESTER AND RAYON

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

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